

Returns to Public vs. Private College: RD Evidence from Centralized Admissions

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Using large administrative datasets from Taiwan, we employ a stacked regression discontinuity design to estimate within each college major the wage returns for students who are marginally admitted into a more prestigious national versus a private college. In these data we observe college applicants admissions outcomes from a centralized system, as well as their eventual earnings. Our results show that at the margin admission into a national versus private college substantively improves college quality and first-year earning outcomes, the latter by about 9 percent or one-fifth standard deviation.

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In most countries of the world, public (or national) colleges are perceived to be categorically the best colleges available. Compared to their private-college counterparts, national colleges are almost uniformly older, larger, and enjoy greater government support in terms of funding and governance; in turn, they attract more and better credentialed faculty and eventually student bodies with higher average abilities. In this way, national-college status is a bundled good, and is an important prestige marker of college quality (Sekhri, 2013).

In this paper, we estimate the monetary returns for students who are on the margin for admissions into either a national or private college within each college major. Our context provides an interesting opportunity to make these college quality comparisons, because the weakest student admitted to the lowest quality national college for any given college major often has scored well within the middle range of the test-score distribution for college entrance exams. (See Figure 2, Panel B.) This suggests that not all national colleges are necessarily elite, yet our results show that even at this level of quality the national college distinction holds value. It also suggests that even average ability students, therefore, still face strong monetary incentives for admission into a national versus a private college.

Our research furthers the current ‘college quality’ literature in at least five areas. First, ours is one of the first papers to use a national universe of microdata for birth outcomes, college applications and admissions, and earnings — all merged on national IDs — to identify causal estimates of the returns to college quality. Second, our reduced-form results employ a regression discontinuity (RD) design that exploits discontinuities generated by the Taiwanese government’s centrally administrated college admissions system; thus, our claim to exogenous variation is highly credible. This comparison at the discontinuities provides an attractive research design because students just above and below an admissions cutoff should be similar in expected outcomes; but because of where these applicants fell in relation to the cutoff, they end up having different college experiences. Conse-

quently, comparing later-life outcomes for these applicants should, on average, provide an estimate of the returns to the differences in the college quality they experienced.

Third, of the known centralized college admissions matching mechanisms used in the world and for which data is available, our data provides the best example in the literature of outcomes from as good as a truthful ranking of preferences as is available. Students during the 2000 admissions seasons were allowed to list 66 preferences¹, which (Kojima and Pathak, 2009) argues from a similar setting, is a sufficiently long list that the incentives to misrepresent their true preferences shrinks to none as the market size enlarges. Thus, the discontinuities we observe are most likely differences in outcomes from best possible matchings.

Fourth, because the data we use catalogs college admissions down to the academic department level, our estimates account not only for the choice of college, but also for the choice of college major — something which had previously been the purview of structural estimation strategies. (For ease of reference, we use college and academic department interchangeably where clarity is not a concern.) Fifth, in terms of research design, the stacked RD we use estimates the local average treatment effect (LATE) at multiple admissions cutoffs in the test score distribution—one for each college major. In this study, the LATE is the estimate of the causal effect of scoring above the lowest possible cutoff for admission into a national college for that college major, explicitly testing the returns to national versus private college even at the lowest possible standard.

Like other recent papers in the college quality literature (Black and Smith, 2006; Dale and Krueger, 2011), we measure college quality as a function of peer quality, proxied by the average college entrance exam score at that college. (See Panel A Table 1 for list of outcome variables.) We also emphasize prestige or selectivity as a quality marker (Brewer et al., 1999; Chevalier and Conlon, 2003); in particular,

¹noted from official document "Joint Board of College Recruitment Commission: Annual Report 2000. Editor and Publisher: College Entrance Examination Center, Taipei.

for many higher education markets in developing countries, a college's status as a national institution suggests that it is of better quality than private institutions offering the same degrees (Rubinstein and Sekhri, 2011), though this does not exclude all private colleges from being elite in these contexts (Hastings et al., 2013). The amount of subsidy a student receives from the institution directly (Van der Klaauw, 2002; Winston, 1999) is also a marker of quality. The previous research above shows a correlation between peer quality and other proxies of college quality, thus making our proxy a useful summary.

In addition to focusing on college quality, this paper also contributes to the 'better schools' literature. Those studies (Jackson, 2010; Pop-Eleches and Urquiola, 2013) focus on attending better high schools and generally show that attending a higher quality high school leads to higher scores on a college entrance exam; notable exceptions to this finding are Clark (2010) in the UK and Abdulkadiroğlu et al. (2011) in Boston/New York, each of which finds no benefit to attending elite exam schools on exit exam scores.

There exists an important difference between returns-to-quality papers² like Abdulkadiroğlu et al. (2011) and Clark (2010) on the one side and Jackson (2010) and Pop-Eleches and Urquiola (2013) on the other: that is, Abdulkadiroğlu et al. (2011) and Clark (2010) include only a very small subset of high-ability students at the very top high schools in their samples, whereas Jackson (2010) in Trinidad-Tobago, Pop-Eleches and Urquiola (2013) in Romania, and Hastings et al. (2013) in Chile look at the quality distribution of all students in the whole market in each of their countries³. Therefore, we may be able to argue that the null returns that are common among the former set of papers can be explained by where the researchers are looking in the student-ability distribution. The latter two examples, using a much broader range of student ability, do show that school quality differences matter. Consequently, the policy interpretation may be that quality

²Rubinstein and Sekhri (2011) and Sekhri (2013) are examples for college quality.

³Hastings et al. (2013) do limit themselves to a estimation sample that includes only oversubscribed schools, but it still covers a wide range of quality.

may have its greatest effects at lower points in the student-ability distribution.

Our paper shares a common focus with papers looking at effects of quality at wide ranges of the quality distribution, instead of focusing only at discontinuities near the top (Saavedra, 2008; Hockstra, 2009). Moreover, unlike papers in the ‘better schools’ literature, we can estimate wage returns in the labor market. And finally, unlike any published paper in the ‘college quality’ or ‘better schools’ literatures, we can do this using all college-goers in a college major and in a cohort.

The remainder of the paper is organized as follows: Section 1 discusses the educational system in the Taiwan context; Section 2 details and summarizes the administrative data sets used. Section 3 motivates the use of RD estimation as a source of identification; Section 4 presents results. Section 5 discusses and Section 6 concludes.

I. General context in Taiwan

Taiwan has a population of roughly 20 million and one of the highest rate of college attendance in the world. By constitutional right, the Ministry of Education (MOE) centrally controls every aspect of Taiwan’s higher education system, as it does at every level.⁴ One of the historically distinctive characteristics of that centralized control is its constraint of supply at every level of education at both public and private schools. This control extends down to each college and academic department in any given year, determining the curriculum that is taught, the number of students and teachers that are allowed in, and even the tuition fees that can be charged.

National colleges are governed by the laws of the University Act⁵ and private

⁴Primary and secondary schools are publicly subsidized and heavily regulated. Junior high school students compete for admissions to either academic or vocational high schools. Within academic high schools, students are tracked into social science versus science subject orientations, with each track preparing students differently to compete in the national exams for their respective track at the college level.

⁵See <http://english.moe.gov.tw/content.asp?CuItem=8173&mp=1>.

colleges by the Private Schools Act⁶. The three most salient differences that appear to create differential value between public and private colleges are that 1) public colleges are fully funded by the government, and operate fiscally independent of tuition revenues, whereas private colleges operate almost exclusively off tuition fees; 2) since the MOE caps all tuition fees, the production function of public versus private colleges operate differently; and 3) though in both cases public and private colleges must have all the policies and procedures approved by the MOE, the supervisory boards that generate these policies and procedures are within the MOE for public colleges and within the college itself for private colleges; thus, these boards have different synergies and constituents. Consequently, public colleges consistently are free to act like non-profits, whereas private colleges have a mix of incentives to work like not-for- and for-profits. Our research estimates the differential impact these combined differences in governance, financing, and operations have on student outcomes.

Besides the historically limited supply, the other distinctive characteristic of Taiwan's higher education market is constrained access, where admissions is determined by performance on a competitive examination and is centrally administered through a national, market-clearing student allocation system. Constrained access has received comparatively less attention in the empirical literature, mainly for lack of good microdata on the appropriate setting needed to facilitate such research. We attempt to fill this gap below.

All college admissions in Taiwan use nationally standardized tests and clear protocols for admissions. Both public and private institutions are subject to the centralized testing regime and the admissions protocol in Taiwan. Students applying to college in the year 2000 have no other option for college enrollment than to follow Taiwan's prescribed protocol. There are two rounds of admissions, early and regular. Figure 1 provides a general sequence of events for early and regular admissions phases in the calendar year. Because these rounds are sepa-

⁶See <http://140.111.112/content.asp?CuItem=8174&mp=1>.

rate in that students cannot compete in both simultaneously, we treat the latter round of regular admissions independent, conditional on early admissions having finished. Regular admissions is the focus of the paper and provides our source of identification. Thus, it is discussed in detail below.⁷

In 2000, approximately 111,150 students applied to college through regular admissions, of which 67,898 or 61 percent were admitted. Of all students admitted to college in Taiwan (74,937), about 91 percent do so through regular admissions.⁸ Students are not allowed to apply directly to colleges, nor can colleges recruit students. Instead, regular admissions is based on students' scores from the Joint College Entrance Exam (hereafter JCEE) — a once-a-year, national standardized battery of tests. In the first week of July each year, any high school senior who hopes to attend college sits for the JCEE. The score on the college entrance exam T_i is a composite of subject-specific tests t_i . The number of subject tests a student must take ranges from 5 to 6 (max score for each is 100 points — not percentiles), and is predetermined by the MOE for each educational tracks: social sciences versus sciences.⁹ Chinese and English are the only two subjects common to all the educational tracks. The scores on the JCEE T_i are a continuous measure, observed down to the hundredth of a decimal point with many data points occurring between any given test-point interval.

After observing her performance on the JCEE, the student must fill in and return to the MOE an 'aspirations card' that lists in ranked order her preferences for specific academic departments within colleges. Concurrent with the context of our data, the maximum number of academic departments which can be listed is 66. A student is only eligible for admissions at departments that are within the

⁷For a thorough discussion of early admissions see the Online Appendix; Simpson and Liu (2013b) uses early admissions as a separate source of variation.

⁸Early admissions was introduced in 1994 and has steadily expanded in the shares of applicants participating and applicants accepted. In 2000, 57,820 high school seniors applied to college through early admissions (47 percent of all college applicants), of which just over 7039 or 13.5 percent were admitted.

⁹According to the MOE's system of categorizing college majors and academic departments that teach them, there are 8 fields of study, which can be further broken down into disciplines and subfields or college majors. Academic departments are categorized under the college majors they offer (www.edu.tw/files/site_content/B0013/bcode.xls).

educational track she has chosen to pursue. Thus, the first rule of admissions is that a student must take the set of subject tests required for her desired college major. For example, if a student chooses to take the required subject tests for the science track, but not the tests for social sciences, she will not be considered by the matching algorithm for any department offering a social-science college major.¹⁰

We do not observe the students' lists of preference, only the department into which they were eventually admitted. The discontinuities that we exploit are at the academic department level. Nevertheless, because of the research design we follow, we compare the quality of departments within a given college major, under the assumption that students who were admitted, for example, into an engineering department at National Taiwan University would have also listed lesser engineering departments at other colleges within a certain range of quality.¹¹ Since for each college major our discontinuity is capturing the difference between one college's offering of that major versus another college's, this means that our analysis reduces again down to a comparison of college quality, taking into account college major. Because students are able to list preferences for academic departments in more than one college major, our results should be interpreted as a relevant estimate, based on one set of possible comparisons.

The matching algorithm (hereafter 'the algorithm') used by the MOE is a school-proposing Deferred Acceptance algorithm. The algorithm does not allow departments and students to choose each other directly, though it does allow for student preferences and department priorities to be taken into account during the matching process. To use the language of Abdulkadiroğlu and Sönmez (2003), departments have a set of "priorities" over student qualities they would like to enroll, whereas students have preferences for specific departments. De-

¹⁰In general, students are already streamed into science versus non-science tracks during admissions into high school. The outcome of their high school admissions conditions the academic track they pursue in college admissions, as the separate high school tracks have differing curriculum to prepare students specifically for the battery of exams corresponding to the track that they are in during high school.

¹¹This range of quality should fall within our widest bandwidth of inspection, 160 points either side of the cutoff, which is about 2.5 s.d. of the running variable.

partment priorities are more abstract in that they are not for specific students, but rather for qualities that any applicant may have. The matching algorithm uses the departments' priorities and students' preferences to match students to seats in specific departments.

The process of the algorithm is as follows:

- 1) The algorithm proposes for each department in each round to the department's current top choice of student(s).
- 2) The algorithm automatically rejects proposals coming from departments where the student has not expressed a preference for the department.
- 3) The algorithm observes for each student in each round any offers from departments the students have preferences for and compares it to the offer(s) they previously accepted, always accepting tentatively the offer from the students' most preferred departments and rejecting the others.

This process continues until all seats in departments are matched to students who had at least some preference for them. The outcome of the matching algorithm is final. Departments are required to admit all those that the algorithm assigns them; there are no wait-lists during the admissions process nor are there transfers between programs after admissions is over. The only way to be admitted into a department is to go through the MOE defined system.

II. Data

This paper uses national administrative data on birth, education, and earnings outcomes for the 2000 cohort of college applicants in Taiwan. All three data sets (birth, education, labor) contain the universe of micro-data for each cohort and are merged using national IDs. Table 1 provides the descriptive statistics. Panel A lists the main outcome variables of interest. Panel B presents other covariates key to the research design. Each of these variables is presented in greater detail below, where each of the three national data sets are described.

A. Admissions data

Because the focus is on college quality, in this paper we use data for only those students who were admitted to college at all. (The returns to a college education is another future project.) Moreover, this paper is using only students admitted via regular admissions, where these specific discontinuities are generated.¹²

The admissions data from the 2000 application season (for students admitted to college for the 2000-2001 academic year) contains all MOE records, including JCEE scores for each of the subject tests t that the applicant i took for either early or regular admissions as well as who applied and was accepted during each round. We use the names of the college and department where the student was admitted to generate two key outcome variables and to construct one of the major contributions of our research design. The two outcome variables are national-college status and department-level tuition. Because every national college has this aspect clearly stated in the college name, we can use this to construct the variable "Pr(National)" is a 0/1 indicator for whether a student was admitted into a national college versus a private college. We can consider the "Pr(National)" indicator to be a proof-of-concept measure for how sharp our RD is working. Second, (natural) "Log Tuition" is constructed by merging the name of the department with MOE records for what each department in the each college was allowed to charge in tuition for that specific year. Lastly, we use the academic department's MOE college-major code to combine all academic departments teaching the same college major into one common running variable. This is crucial to our RD being about to account for both choice of college and college major.

From these individual subject-specific test scores we have for each student, we calculate the variable "Peer Quality @ College" is the college quality proxied by the average JCEE score at the college where the student was admitted. The specific set of subject-tests a student is required to take is predetermined by the

¹²In another project, Simpson and Liu (2013b) deals with the returns to college quality using outcomes and the source of exogenous variation created from early admissions.

MOE. Thus, depending on the educational track a student pursues, students could earn a maximum of either 500 or 600 points on their set of required tests for the JCEE. Totaled test scores in the data range from 134.19 to 578.17.

To capture a common measure for the performance of students in same college but different educational tracks, we use the percentage of total possible scores:¹³ this is calculated by dividing the individual scores by the total possible (either 500 or 600, depending on what was required by the MOE) and then multiplying by 100. From this, peer quality (denoted as $\bar{A}_i$ in the formula below) can be calculated by averaging the percentages at either college- or department-levels. The value of n in the formula's denominator changes depending on whether the college or department level is used.

$$\bar{A}_i = \frac{\sum_{i=1}^n \left[\left(\frac{\text{actual score}_i}{\text{total possible score}_i} \right) * 100 \right]}{n}$$

Because this study focuses on the returns to attending a national college versus a private college for the same college major, we use only those college majors that have both national and private colleges offering it. Some college majors have no national colleges teaching it. Conversely, during our time period, some college majors were only found in national colleges, for example, education. Additionally, some private colleges have (top) enrollees who have test scores so far from the cutoff that when working even within a wide bandwidth, none of the students who were admitted into these private colleges fell within even 80 points¹⁴ of the admission cut off for the national college with the lowest rank. In all three cases, it is impossible to compare the relative advantages of attending a national college versus not when there is no variation in national college status occurring within that college major or within the bandwidth of analysis. Thus, these college majors were explicitly dropped from the analysis. The result is that college majors in the

¹³Note that the points described here are in terms of the UNWEIGHTED (or raw) test scores on the JCEE for each department, which is distinct from the WEIGHTED test scores used to make the running variable in the RD, described more fully below.

¹⁴Almost 1.5 standard deviations of the running variable distribution of scores

following fields of study were completely dropped from the analysis: education, humanities and arts, and service-industry. The remaining college majors consist of some of the most common and/or competitive: e.g., business & law from social sciences and math, engineer, physical & agricultural sciences. Chay et al. (2005) and Hastings et al. (2013) in their studies on Chilean data, for example, each similarly limit their working samples to those data which occur around identifiable discontinuities. Our data includes admissions into programs at 50 colleges: 21 of which are national colleges, the remaining 29 are private.

B. Birth certificate data

We are able to match 100 percent of the individuals in the MOE data to their birth certificate data using the universe of all births in Taiwan for the birth cohorts 1978-1983. The birth data are collected by the Ministry of the Interior Affairs. Of the data collected on the birth certificate, we use the following variables: year and month of birth from which age is calculated, child's sex from which we construct the male 0/1 indicator, and the parental years of schooling for each student at the time of his/her birth. (See Table 1, Panel B.)

C. Earnings data

The administrative data we use on earnings is for the years 2004 to 2005; 2004 is the first year graduates would have been able to find a full-time job. The data is administered by Taiwan's Bureau of Labor Insurance and collected by the Council of Labor Affairs. It is an employer-employee matched data set, which includes the employee's national ID and monthly earnings for that year. (See Figure 3.)

The Labor Insurance data (hereafter "the Labor data") is a record of healthcare insurance provided through employers. All employers are required by law to submit monthly earnings records for their employees in order for the monthly premium to be removed from their monthly earnings (Chou et al., 2003). Because of the bureaucratic process for setting the healthcare premia with its floor, ceiling,

and menu pricing, the maximum wage in our records is 42,000 NTD, which is sufficiently high for the earnings of most new graduates. Any wage above 42,000 was top-coded at the upper limit by the Bureau of Labor Insurance. Similarly, the minimum wage is 11,000 NTD, which again is well below the average earnings. Any wage below 11,000 was bottom-coded at the lower limit.¹⁵ The wages are measured at a fairly fine-grained interval of about 600 NTDs. The data contain the complete universe of all private-sector workers in Taiwan, approximately 82 percent of the working population. Of those working adults not in the Labor data, about 10 percent are public-sector employees, 3 percent are farmers/fishermen, and 5 percent are self-employed. The former three maintain a different form of public insurance that pre-dated universal healthcare. All unemployed individuals (unemployment rate is about 3-4 percent at this time) are also in this data ever since 1995 (Chou et al., 2003); however, they of course will not have wages.

A significant limitation of our study is that the Labor data to which we have access can be matched to only about 33 percent of those in our MOE records. This could be due to multiple reasons consistent with the average state of the labor market at this time. The main reason for the lower match rate most likely is the 18-month national military service that is mandatory for males at this time — unless they were exempted or able to defer their conscription.¹⁶ An 18-month military service starting in June 2004 would end in December 2005; thus, the earliest we could expect these students to be employed would be January 2006. There are twice the number of females than males matched with their wages in the overall dataset. Reasons why a female graduate is not in our data are most likely due to their being disproportionately employed in education or other government jobs, which is more stable and less competitive. This reason for females to be

¹⁵We ran Tobit regressions on the data to account for the censored nature of the top- and bottom-coding. The results are the same; therefore, we present all results in OLS only.

¹⁶Reasons for not performing the service immediately (or at all) following graduation from college would be continued education at the post-graduate level, being deemed physically unfit for various reasons, being the only child and thus would leave the parents without a child to assist them, having another family member currently servicing in the military, or fulfilling a bond with a government research position.

missing is further supported by the fact that once we narrow our research sample down to college majors which meet our criteria (listed in Section 2.1) for having both national and private colleges, the share of males increases to around 62 percent of the sample. Unfortunately, access to the Labor data for 2006 is not currently possible.

As mentioned above, a second reason for the lower match rate is that about 10 percent of all working adults hold a position in the government, which could include bureaucrats, public school teachers, or public service workers, to name a few. These public servants would not appear in the Labor data. Other reasons explaining the lower match rate with wages are that graduates went on to graduate school (though this is even more restricted access than college in Taiwan) or they are unemployed (the unemployment rate is around 4 percent for college and university graduates at this time) (Yang et al., 2010). With our data we are unable to distinguish between these different states.

To determine whether this data limitation signals some element of selection, we check for balance in the probability of having wages on either side of the discontinuity. Panel A of Figure 4 shows that there is balance in probability of showing up in the Labor data on either side of the discontinuity. Thus, selection into employment and sector do not appear to be driving our reduced-form results. Further details regarding the Labor data are discussed in the Data Appendix.

III. Empirical Strategy

Our identification strategy follows the basic regression discontinuity design: a student's weighted exam score is the running or forcing variable (x-axis) in the RD. A student's score is either above or below the cutscore c for admissions into that department.¹⁷ The RD design works in the spirit of a randomized experiment in that those who are just above and just below the admissions threshold are

¹⁷In previous versions of the paper, we constructed the running variable using the student's rank or priority ordering at each department instead of their weighted test score for that department. The basic results from this design remain consistent with our findings here.

essentially the same in observable characteristics, most obviously in their nearly identical test scores. In many cases where admissions cutoffs are set based on test scores, the random measurement error inherent to the construction of standardized tests provides enough random variation to support the claim that students on either side of the cutoff are identical *in expectation*. Thus, those who are just below the threshold provide a good comparison group for those who are just above, and the differences in outcomes that occur later are plausibly due to the realized differences in the college quality they experience.

A. RD with Single Discontinuity

Consider a hypothetical case in which there are two departments d_1 and d_2 , ranked from lowest to highest by the quality of the department. Department d_2 has more applicants than there is capacity. One cutoff c_1 defines the access threshold to d_2 , the department with the higher quality. Students scoring above c_1 are eligible to attend d_2 .

Let the following model (1) represent admissions outcomes — at both the first-stage and reduced-form stages — as a function of that student's achievement on the college entrance exam T_{id} ¹⁸:

$$(1) \quad Y_i = \beta 1(T_i \geq c_d) + f(T_i - c_d) + 1(T_i \geq c_d) * f(T_i - c_d) + \mathbf{X}_i + \varepsilon_i$$

where $1(T_{id} \geq c_d)$ is an indicator function for whether a student's total weighted test score T_{id} is above the cutoff c_d for access to d . The function $f(T_{id} - c_d)$ is the summation of higher-order polynomials of order p , independently flexible above and below the cutscore. Within a narrow enough bandwidth around the cutoff c_1 (in this hypothetical case), a flexible specification of $f(T_{id} - c_d)$ fully controls for other factors driving access to d_2 . At widest bandwidths in the running variable

¹⁸We follow the MOE's process for generating weighted test scores T_{id} , as is described in the Online Appendix under "How Colleges Rank Students."

we use a 6th-order polynomial, then 5th-, 4th- and 3rd-order as we narrow the bandwidth; and at the narrowest bandwidths we use quadratic and linear fits. The error term ε_i is clustered at the department level, taking into account that students who applied to the same departments may have similar errors in the reporting of their record-keeping and/or some unobserved similarity.

We use three proxies for college quality: the probability of being admitted into a national college $Pr(National)$, the average peer quality $\overline{A}_i$ at the college-level, and natural log of yearly tuition $ln(Tuition)$. Whether or not a student has a degree from a national college is an important label of success in Taiwan, as it is in other non-Western contexts (Rubinstein and Sekhri, 2011). Thus, we use $Pr(National)$ as an indicator for the college-prestige signal employers would observe when looking at an applicant's resume.¹⁹ A finer-grained measure of prestige (but one that is harder to observe with precision by the potential employer) is the average peer quality of the institution, proxied by $\overline{A}_i$. Lastly, another valuable feature of attending a national college is that it is subsidized by the government, and thus would cost less than what private colleges offering the same major would cost (Liu et al., 2006).

In the first-stage analyses, we interpret β as the expected improvement in college quality (in term of our three proxies) that a student experiences by scoring above the cutoff for access to d_2 . Because we are only considering the differentials at the discontinuity associated with admissions to (at least) the lowest quality national college for that college major, we interpret each of these outcomes as the benefits for admissions into a more prestigious college.

Now re-consider model (1) with the outcome log-earnings Y_i again a function of a student's performance on the college entrance exam. The β coefficient on the term $1(T_{id} \geq c_1)$ now represents the increase in log-earnings (log-points) one is expected to receive from scoring just above the cutoff compared to just below. Thus, this is the monetary return to attending a national college versus a private

¹⁹It also can be interpreted as a measure of how fuzzy our discontinuity is.

college, conditional on college major and (implicitly) college admission at all.

Notice that the outcome Y_i in model (1) is fundamentally a function of the relative distance a student's test score is from the cutoff. The intention-to-treat (ITT) interpretation of the β coefficient represents the reduced-form change in some outcome Y_i (whether academic or labor market) that i experiences from scoring just above the cutscore compared to just below. A vector of student characteristics X_i are included, but are not required for the β coefficient to be causally identified.

An identifying assumption of the RD design is that students are not able to manipulate their relative position to the cutoff. The rigors of the matching algorithm itself should eliminate this. Moreover, the fact that students can list up to 80 preferences on their "aspirations card" means that their rankings should be as good as truthful preferences (Kojima and Pathak, 2009). Two ways in which one could test for this is to check the density of the data around the cutoff and to see if, on either side of the cutoff, the background characteristics are otherwise similar. Figure 2 Panel B presents the density of the running variable, that is, the weighted recentered test scores. The density corresponds closely to that of Panel A in Figure 2 (the raw scores), which in turn approximates a normal distribution, as one might expect from such a process as regular admissions. Though it is fundamentally impossible to check whether all student characteristics are balanced, we test for balance in the characteristics we have access to by inspecting whether there is overlap in their confidence intervals on either side of the discontinuity. In Figure 4 Panels B-E we see that the confidence intervals for each of these covariates do indeed overlap, indicating that the students are similar across the cutscore. From these results, we can infer that the treated and control samples are equivalent, and thus the positive effects we observe in the outcomes of interest are not due to substantive differences in student characteristics around the cutoff.

B. Stacked RD with Multiple Discontinuities

Our analysis exploits 35 discontinuities for access into a national college. Similar to Pop-Eleches and Urquiola (2013), we use a stacked RD set up. For each college major, scoring about the admissions cutoff to a national college is treated as a separate RD — a self-contained natural experiment with its own LATE estimate. By stacking all the RDs, we can aggregate all the LATEs to provide one estimate of the impact. Besides simplifying the task of interpreting the results, stacking RDs has the desirable characteristics of increasing statistical power by pooling observations. Stacking entails re-centering the running variable T_{id} for each student i . We do this by normalizing all student test scores around the minimum cutoff c_d required for admission to the lowest-ranked national college for that college major: $T_{id} - c_d$. We normalize test scores within college majors; thus, the subscript d in $T_{id} - c_d$ refers only to the academic department of lowest-ranked national college for that college major.

When stacking RDs, it is common to include a fixed effect γ_c to take into account that multiple RDs are being combined. Because the first-stage outcomes only vary at the department level; including a department-level fixed effect in these regressions would absorb all the variation; therefore, our main specification includes cutoff c_d fixed effects, which also take into account that the cutoffs occur at different points in the distribution of T . However, for robustness checks on our wage results (which variables at the individual level) we also experiment with other fixed-effect specifications by including fixed effects for each department d and also for each college major m . Model (2) incorporates the fixed effect specification.

$$(2) \quad Y_i = \beta 1(T_i \geq c_d) + f(T_i - c_d) + 1(T_i \geq c_d) * f(T_i - c_d) + \gamma(\mathbf{c})_d + \mathbf{X}_i + \varepsilon_i$$

Because re-centering and stacking the data multiple times aligns the discontinuities for each department at the cutoff, it increases the probability that there

will always be an observation at $T=0$ and, therefore, mechanically increases the density of re-centered test scores at the cutoff. This artificial peak has nothing to do with students being able to manipulate which side of the cutoff they land on. Following Pop-Eleches and Urquiola (2013)'s example, we run a 'donut-RD' by dropping observations at $T=0$.

IV. Results

To summarize our results, scoring above the cutoff provides applicants with 50 percent greater chance of getting into a more prestigious college and have access to peers of 0.2σ higher average ability. This quality improvement translates into a wage differential of around 7-11 log-points (on a 2-digit log base) or 0.22σ in log earnings or about 9-10 percent more each month.

Our preferred estimation strategy is to cross-validate the results on a wide range of specifications. The basic model fits the data starting with a linear fit at a bandwidth of 20, then increases the polynomial fit incrementally from quadratic, to cubic, quartic, all the way up to 6th-order at the widest bandwidths, as is also suggested by (Imbens and Lemieux, 2008).

Table 2 provides the main results. The tabular results match well with what we see graphically in Figures 5, 6, and 8—with each figure presenting residuals from different bandwidths and smoothing functions (linear or polynomial).²⁰ Panel A in each figure presents results using a linear fit and bandwidths of 20 points, respectively, around the cutoff. Panels B and C present results using a polynomial fit and bandwidths of 40 and 60 points, respectively, around the cutoff. All residuals are from a regression of the outcome variable (first-stage and reduced-form) onto a model including a linear trend in re-centered test scores, the covariates, and cutoff fixed effects. Figure 9 presents all results in terms of raw data and residual.

²⁰Graphical analysis are not provided for log-tuition results as they are the exact inversion of those for Figure 5. However, they are provided in Figure 9.

First-stage: Figure 5 and Panel A of Table 2 (Columns 1-7) provide the cross-validated results for impacts of scoring above the cutoff on national college admittance. They look quite similar in size and significance across different bandwidths and specifications. These results in Table 2, Panel A show that scoring just above the cutoff predicts that a student has a 30-35 percentage points more likely to enter a national college; this is an increase of about 50 percent over the base. These are sizable differentials (0.6σ) in prestige returns for a less-than one-point difference in JCEE admissions score. 

Since this is not a sharp RD with 0-to-1 at the discontinuity, it also suggests that some students have ranked a private college more highly than a national one. Such factors as distance to college or even where classmates have chosen to go, could explain this. To some extent we would expect there to be mismeasurement, since we lack students' actual listing of preferences. Nevertheless, treating our results as product of a fuzzy RD, even with mismeasurement, we still observe stark and sizable discontinuity. This 30-35 percentage point increase in national status would further suggest that, if we so desired, we could scale the effects we observe below by an approximate factor of 3.

Figures 6 (Panel B in Table 2) shows the impact on peer quality of scoring just above versus below the admissions cutoff for a national college in your college major. This specific outcome is more sensitive to functional form and bandwidth; this sensitivity to functional form can be seen most notably in Figure 6, where in Panel A a linear specification approximates the data but a polynomial one improves the fit immensely in Panels B and C. The improved model fit sometimes doubles the magnitude and trebles the statistical significance of the coefficient at discontinuity to greater than 2 percentage-point increase in peer quality, as we can see in Columns 8 and 9 of Table 2. The interpretation of the jump at the discontinuity is that for those scoring above the cutoff, one's classmates have on average achieved 2 percent points higher on the college entrance exam. Though this may initially appear to be only a small improvement, when compared against

the variance in the whole sample (10 percentage points), the 2 percentage-point increase is just under one-fifth s.d. (0.2σ).

Because the tuition a student pays each year is (for each college major) a function of whether or not she attended a national or private college, then it is unsurprising that Panel C of Table 2 is approximately the inverse of Panel A of Table in each of the figures, as can be vividly seen by comparing Panels A and C in Figure 9. Students who attend national colleges can pay up to as much as one-half the tuition that private students pay for the same college degree in the same college major (Liu et al., 2006). This is because the government heavily subsidizes national colleges. The negative coefficient in the natural log of tuition show around 20 log points *reduction*, or about 22 percent *decrease* at the margin. That is, students who score just above the admissions cutoff for a national college are paying over 8000 NTDs less in tuition. Like its counterpart in probability of being admitted into a national college, the discontinuity in log tuition is equivalent to a 0.6σ tuition subsidy.

All three of these impacts suggest that admission into a national college carries with it a variety of benefits experienced during the college-going years. These first-stage effects are likely to have several non-monetary and non-cognitive benefits for which we have no measure. For example, anecdotal evidence from around the world (but especially in East Asia) suggests that the prestige of the school where one's child attends is of major consumption value to parents. Thus, there is much pressure for children to succeed in this area. Being admitted into a prestigious college 'gives face' to parents by possibly vindicating their parenting skills or genetic pedigree. It also likely increases confidence in the student's own ability and future successes. In the next section we explore evidence that these first-stage benefits also have reduced-form impacts on monetary returns.

Reduced-form: The first-stage impacts shown above correspond to sizable reduced-form impacts on log wages. Table 2 Panel D (Columns 22-28) shows that scoring

just above the cutoff for admission into a national college for a college major improved log wages anywhere between 7 to 11 log-points, or about 0.22σ in log wages. That is about 1600 to 1700 NTDs or 8 percent of average monthly earnings for this subpopulation of early graduates. As with the graphical results for peer quality, the high-order polynomials provide a much better fit as the bandwidth increases. This can best be seen by comparing Panels A and B in Figure 8, or from a linear fit to a polynomial one.

Table 3 shows wage results from different fixed-effect specifications. Panel A provides estimates with fixed effects at the department level. Letting each department have its own intercept absorbs all other variation above and below the cutoff except for a college's national versus private status. Across the specifications, the coefficient increase in size by about one-third to one-half and in significance as well over their counterparts in Table 2 Panel D. Panel B of Table 3 replaces department-level fixed effects for college major fixed effects; since each college major had a different cut-off, these consequently are the same as Table 2, Panel D.

Table 3 in Panel C offers the results from a random-slopes model. In addition to college-major fixed effects, we now interact the each fixed effect with the running variable above and below the cutoff. This specification is stricter on the data and, thus, is a more robust test of the model since it allows the running variable for college major to have its own intercept and slope around the cutoff. Such a specification more flexibly controls for variation in the outcome. Overall, these three specifications present consistent findings and make a robust for the admissions cutoff creating quality difference that then create differences in wages.²¹

Robustness checks: Tables 4 present cross-validated results (peer quality and

²¹Specifications from previous versions of the paper use a running variable constructed on the normalized ranking of students' priority orderings at each department. They provide qualitatively similar results.

log wage) with the data split into social- versus science-major subsamples.^{22,23} Though we have chosen to cross-validate on the widest specifications and 5th-degree polynomials, the pattern of results are consistent on more narrow bandwidths and lower-order specifications.

Looking at Panel A of Table 4 we can see that the odd-numbered columns are for the social-majors and the even-numbered columns are for science-majors. Splitting the data this way gives a larger share of the observations to the science-majors at each bandwidth.²⁴ However, once we inspect the sample sizes in Panel B by their social- and science-tracks, we see that this proportionality switches, with social-majors being more heavily represented than science-majors in the sample that has wages. Social-majors outweighs science-majors by about twice as much for each bandwidth.

Nevertheless, upon consideration of the coefficients in Table 4 for both Panels A and B, we see that it is consistently the science majors that show both statistically and substantially significant results, whereas the social-majors are both statistically insignificant and substantially smaller in size across each bandwidth. And in Panel A, the sign of the peer quality changes direction as the bandwidth increases from 110 to 140 points. These stark differences in magnitudes in peer quality explain somewhat why we see that the even-column coefficients are larger than the ones we see in Panel B of Table 2, where all the college majors are combined.

Likewise, in Panel B for log wages, though drastically diminished sample size for the sciences is lessening the statistical significance, we still see that science-

²²Social science college majors in the main data set fall into three main disciplines: social sciences majors proper such as economics or sociology, business majors, and law majors. Science college majors fall into 4 main disciplines: math and engineering, physical sciences, agri- and bio-sciences, and medical sciences.

²³Each of these disciplines contain separate college majors within their categorization. Splitting the data into smaller subsets is technically possible, but the sample sizes would shrink dramatically, and thus power issues would prevail.

²⁴Remember that college majors that were dropped from this study (because there was not the necessary variation in private versus national colleges offering that college major) fell into 3 social-science fields of study. This gives the sciences a slightly higher representation in this analysis than is characteristic of the population as a whole.

majors are experiencing robust returns for scoring above the cutoff, while the social-majors are not. These results are even larger in magnitude as well, around 11 to 14 log-points or about 12 percent more each month (around 2,000 NTD or around 0.25σ).

Tables 5 present cross-validated results (peer quality and log wage) with the data split into female versus male subsamples. These results are less easily interpreted. In Panel A, the relationship to sub-sample size and gender would suggest that many males study science-majors, though obviously not all. The balance is about 70/30 for males studying science versus social. This balance is slightly less skewed at 60/40 in the sample with wages. For Panel A, the peer quality shows that after 110 points in bandwidth around the cutoff, the coefficients for females no longer have any statistical significance, though their magnitude shrink only slightly.

The opposite holds true for Table 5, Panel B where log-wage results are presented. Females now have statistically significant returns at the discontinuity for access to a national college, whereas males do not—though throughout the coefficient stay positive and around the same general size in magnitude. Several things may be happening in these regressions, but clearly an underlying interaction between gender and track-of-study is effecting the results.

To pursue further this connection between gender and track-of-study, in the regressions for Table 6, we again use the combined sample and include an interaction between gender and track-of-study in the model. The base term for track of study (science vs. not) for this model is captured much more flexibly by the college-major fixed effects and their random slope effects. The new gender-track interaction term allows us to absorb the imprecision in the data. Indeed, adding this term significantly improves the precision of the coefficients in both Panels A and B such that they remain statistically significant and substantively stable. The first-stage impact is about a 1.5 percentage-point increase in peer quality and corresponding reduced-form is 9 log-point increase in wages across all the

bandwidths and polynomial fits. These results are our most stringent and robust results for the effect of college quality on wages.

V. Discussion

We show above compelling evidence that even at the margin admission into the national college for a given college major does convey substantive benefits. The 50 percent increase in national college admissions at the cutoff suggests that we have identified an interesting source of variation in access to these benefits. To provide further evidence that greater government resources follows the distinction between national and private, we show that students who score just above the cutoff for admission into national colleges enjoy a yearly tuition subsidy of around 8000 NTD, compared to those studying the same college major between went to a private college. Since the government caps the amount a tuition colleges can charge, this amounts to about a one-quarter of the yearly tuition or 0.6 σ change. Both of these outcomes appear to follow a very similar pattern that supports the interpretation: significant changes in government resources are occurring at this margin.²⁵

The evidence suggests that, unsurprisingly, students respond to these government investments in higher education. As our findings show, students of higher ability are attracted to national colleges over private colleges. This is happening discretely at the margin as in Panels A-C of Figure 6, where we observe a jump in the average quality of the peers that were admitted at these colleges. And yet it is not the same single-discontinuity pattern we saw in the panels of Figure 5. Moving left to right along the x-axes of Panels B and C in Figure 6, we see the peer quality at the college level takes on a zig-zag relationship, especially distinct to the right of the cutoff. The sharp break in direction for these de-trended residuals suggests that there are cutoffs for other levels of college quality, especially among

²⁵Because tuition has much to do with parents' own monetary well-being, in one sense as an outcome it could function somewhat like a second-stage, reduced-form outcome. However, we have treated it here as first-stage evidence for a general claim that national colleges enjoy greater government support.

national colleges. This is consistent with the contextual fact that among the national colleges, there is an elite of five or six—with National Taiwan University being the top.

Figure 9, Panels A, B, and D reproduces these graphs from Figures 5, 6, and 8, which use residuals, and is paired along with the raw data counterparts. The first observation that we can make regarding this comparison is that the residuals allow for greater inspection of some underlying dynamics, especially in peer quality. Nevertheless, even in the raw data (as we see in the second-row left-hand graph) for peer quality, a suggestive pattern of ascending steps to the right of the cutoff, which reinforces our explanation that national colleges display a distinct pattern of higher quality at the margin, which is not present among the lower quality private colleges.

Another (possibly more) important observation is that the very distinct zigs and zags we see in the detrended college-level peer quality patterns is not present in the detrended results for log wages. Comparing the raw and residual evidence in Figure 9, Panel D for log wages, we see that they look relatively the same. The kinks in the trend line to the right of the cutoff in peer quality are not present to the right of the cutoff for wages. Using these between- and within-outcome comparisons, we suggest that wage returns may not be as responsive to the college-level variation in peer quality as at first thought.

Probing further, we look back again at Figures 6 and 7, which display college- and department-level variations in peer quality (residuals), respectively. The consistent zig-zag that shows up at the college level in Figure 6 is less obvious when we look at the department level in Figure 7, especially to the right of the cutoff. The tightness of the zigs are still evident, but there are fewer of them—one to be exact. This may suggest that even at the department level there exists a discernible difference in peer quality at the margin for entering an elite top-6 national college. But overall, for both college and department-level variation, the discontinuity at the cutoff for entrance into the lowest quality national college

still appear to be the largest trend breaks. This resonates with our interpretation of earlier studies (Abdulkadiroğlu et al., 2011; Clark, 2010; Jackson, 2010; Pop-Eleches and Urquiola, 2013), that variation in quality occurring at the top of the distribution may not be as large or as easily discernible, whereas quality variation at points towards the middle of the overall distribution is likely to be larger and thus have greater effect.

Our data appears to exhibit a possibly stronger correspondence in the log-wages to the department-level peer quality than to the college-level, especially as we move outward from the cutoff. Most important regarding that underlying correspondence is that we are unlikely to observe larger discontinuities in wages at other admissions cutoffs within the same college major, because the discontinuities in peer quality which are so distinctive at the college level are not as distinctive at the department level.

Of course, as our analysis moves away from the cutoff in an RD setup, we begin to move away from the strengths of our causal identification and more into the realm of speculation. Lacking other sources of exogenous variation to independently identifying the mechanism, it remains hard to disentangle exactly what it is about the national-college status²⁶ that employers are rewarding with higher wages, since we observe discontinuous changes in several underlying characteristics that are consistent with national colleges being of higher quality. This is unfortunate because research into college quality (and higher education more broadly) is motivated implicitly by a social-planner's desire to identify technologies, as it were, that could be generalized to other contexts. As the peer quality literature has pointed out, simply moving smart students from one school to make another school better off in peer quality can potentially dissolve down into a zero-sum game. We want transferable resources that are not necessarily resource constrained.

²⁶and the greater bundle of college quality it affords

VI. Conclusion

In this paper we have exploited plausibly exogenous variation as a source of identification for estimating the returns to college quality: Students scoring above the admissions cutoff for the lowest-ranked national college were more likely to be exposed to better college quality as measured by several proxies. We show that students above the cutoff, on average, experienced 0.2 σ better peers at college, were 50 percent (35 percentage points or 0.6 σ) more likely to attend a national college, paid about 22 percent (20 log-points, 0.6 σ , or 8000 NTD) *less* in tuition each year. These students scoring above the admissions cutoff eventually earned around 9-10 percent more (9-10 log-points, 0.20 σ or about 1700-1800 higher wages) more each month in their first year of employment after graduation.

Though this paper cannot completely disentangle the underlying mechanism which cause the higher quality institution and departments to cause higher earnings, we have convincingly established that they do. This is one of the first papers to estimate the causal returns to college quality in which the whole college market of colleges and college students is available for analysis. The evidence above suggests broadly that college quality matters. Aside from being responsive to the bundle of benefits that come from national-college status, wages also appear to follow a similar pattern as the quality of the academic department where they attend.²⁷ The latter finding is not wholly unexpected, especially since the MOE has regular and direct supervision with the funding, staffing, admissions, and curriculum of departments, not just colleges. Further, it could be reasonable to assume that professors, especially those that run their own labs, will have industry connections that are more strongly associated with their own research than necessarily with the college where they are employed.

²⁷Further anecdotal evidence suggests that government resources play a big part in the productivity of colleges. Post-2005 the Taiwanese government has broadened their commitment to funding college quality, independent of national status. In a bid to have a top-100 world university by 2015 the government has committed to offer financial support to any institution that can make competitive proposals. The outcome of such a policy change will likely shift both the productivity of the institution as well as attract smarter students; it is unclear which effect will be larger (Lo, 2010).

The generalizability of these results may be limited to country contexts that are similarly centralized. Though this does exclude countries like the US, it nevertheless includes many other countries that have experienced the largest growth in higher education and, in the case of other East Asian countries, national populations with the world's highest levels of college education. Therefore, this paper offers early and important insights into the causal effects of college quality on earnings.

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Figures



FIGURE 1. TIMELINE OF EARLY AND REGULAR ADMISSIONS.

Notes: This graph is an approximation.

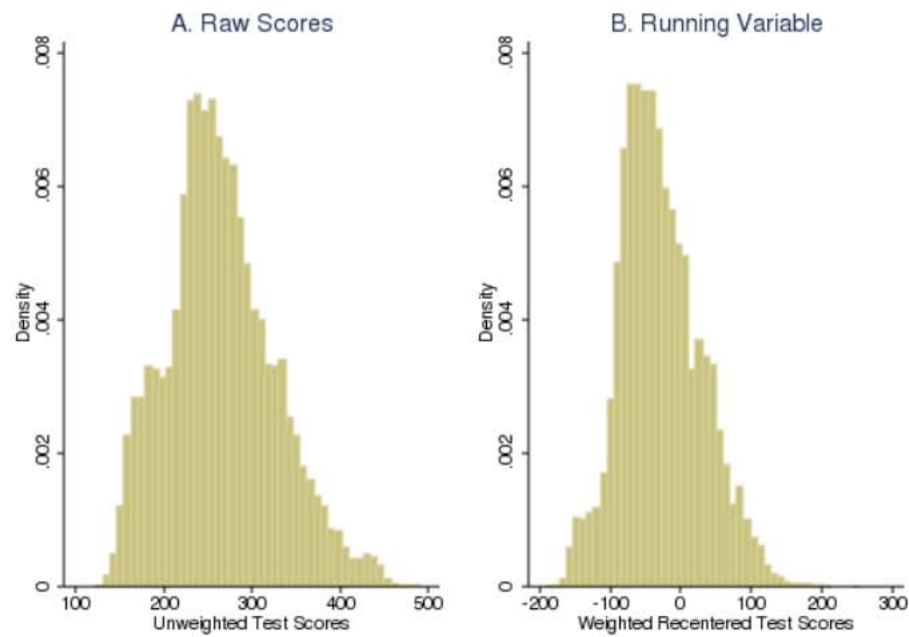


FIGURE 2. HISTOGRAM OF RECENTERED RUNNING VARIABLE

Notes: The plots above presents the density of the test score data. Panel A provides the total summation of unweighted test scores for required subjects where the student was admitted. Panel B provides the total summation of weighted test scores for required subjects where the student was admitted, that were then recentered around the lowest cutoff for the national in each college major.

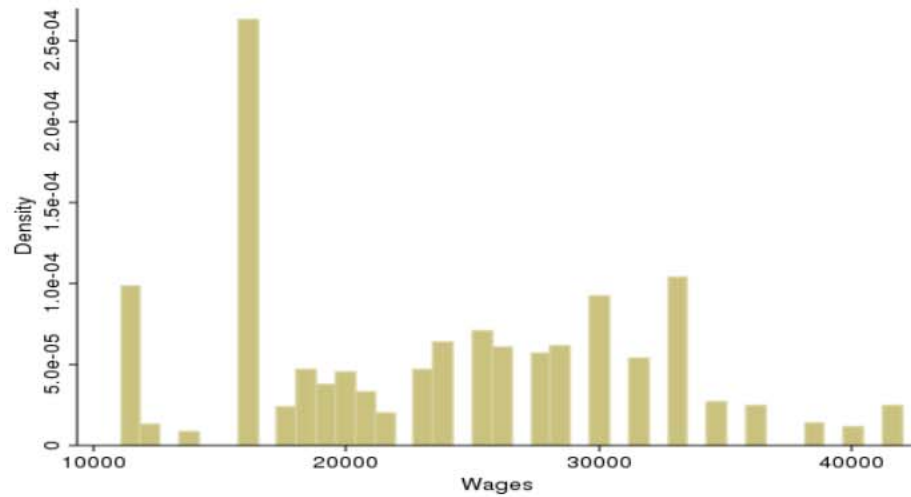


FIGURE 3. HISTOGRAM OF WAGES

Notes: These represent the first year monthly earnings (in New Taiwan Dollars) for those entering college in 2000. The data were recorded in the month of December for 2004 and 2005.

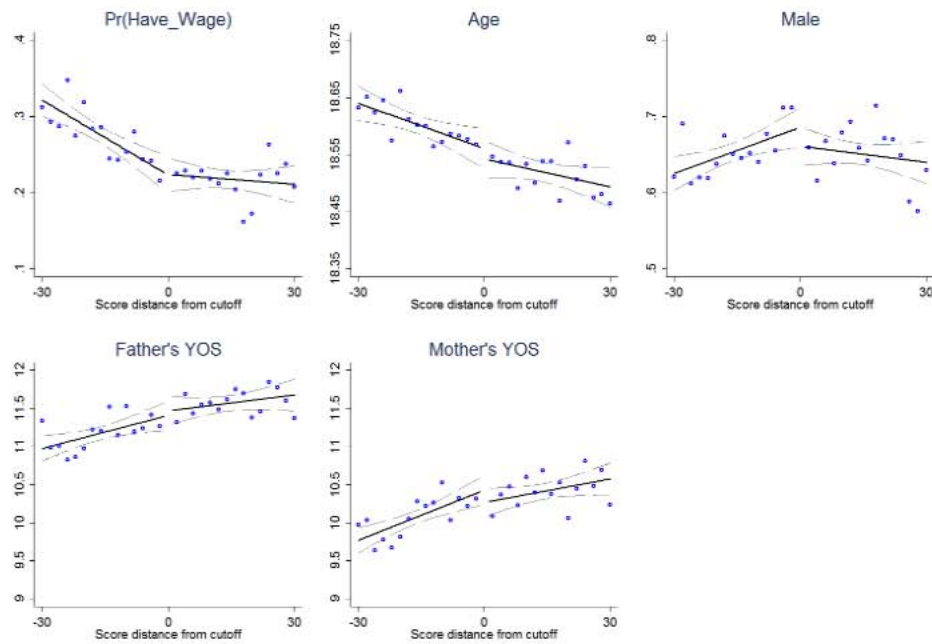


FIGURE 4. BALANCE IN THE COVARIATES

Notes: The plots above present the linearly smoothed fits of each control variable around the admissions cutoff. The dots represent the bin averages using a 2-point bin-width.

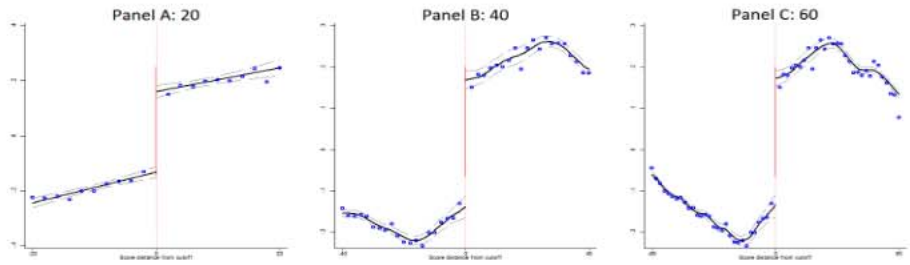


FIGURE 5. Pr(NATIONAL): BANDWIDTHS 20, 40, 60

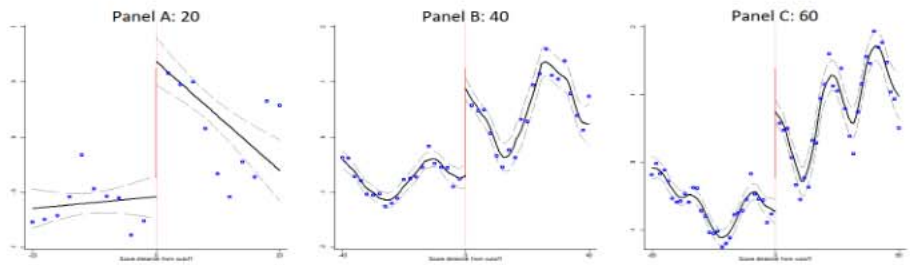


FIGURE 6. PEER QUALITY @ COLLEGE: BANDWIDTHS 20, 40, 60

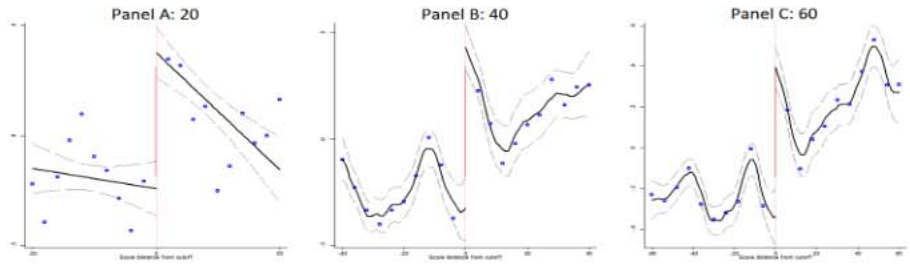


FIGURE 7. PEER QUALITY @ DEPARTMENT: BANDWIDTHS 20, 40, 60

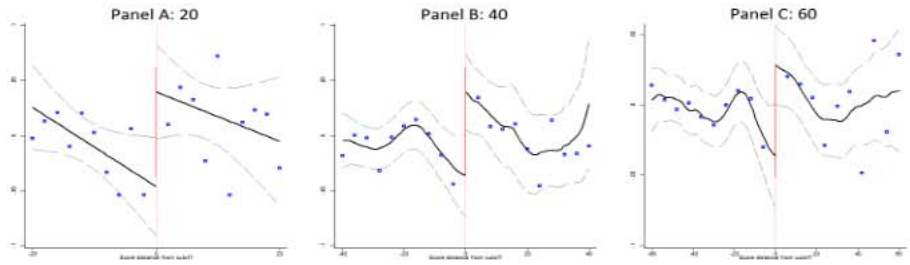


FIGURE 8. LOG WAGES: BANDWIDTHS 20, 40, 60

Notes: The plots above present smoothed fits of residuals from a regression of each outcome variable (first-stage and reduced-form) on to a model including a linear trend in re-centered test scores, the covariates, and cutoff fixed effects. Figure 6 presents the peer quality at the department level, as opposed to the college level as in Figure 7. The effect of log-tuition is the exact inverse of Pr(National), thus not shown here. (See Figure 9, Panel C.) Figure 8 shows log-wages.

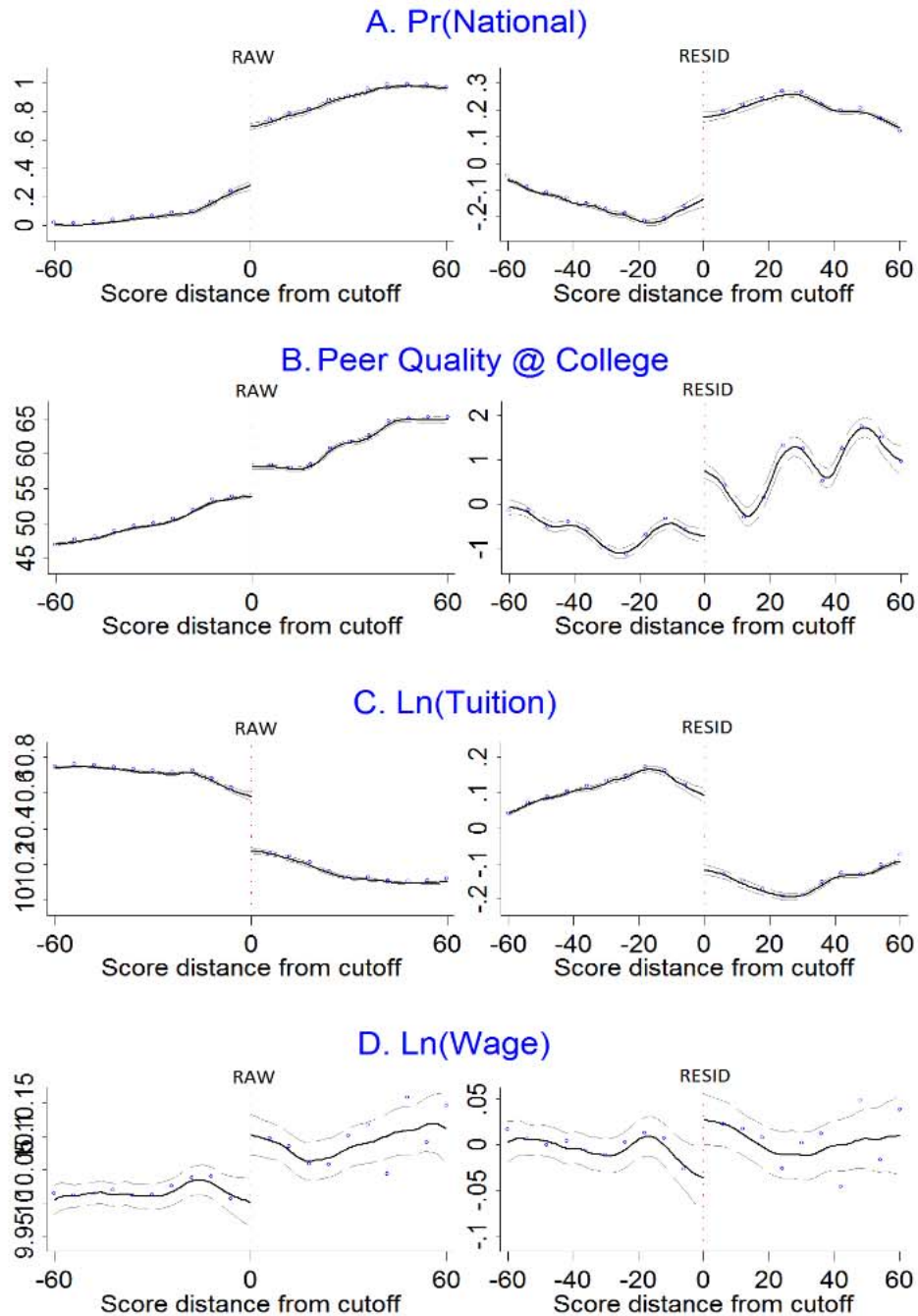


FIGURE 9. COMPARING RESIDUALS AND RAW DATA GRAPHICAL RESULTS

Notes: The plots above present the smoothed polynomial fits of raw data (left panels) as well as residuals (right panels) on 60-point bandwidth. First-stage results are the first 3 panels: “Pr(National)”, “Peer Quality @ College”, and “Ln(Tuition)”. The reduced-form log-wage result is the last panel: “Ln(Wage).”

Tables

TABLE 1—DESCRIPTIVE STATISTICS

Variable	Obs	Mean	Std. Dev.	Min	Max
A. Outcomes					
Pr(National)	39951	0.29	0.45	0	1
Peer Quality @ College	39951	51.63	9.44	34.57	71.28
Log Tuition	39951	10.57	0.32	9.89	11.03
Wage	11255	23681.63	7762.68	11100	42000
Log Wage	11255	10.01	0.35	9.32	10.65
B. Other Variables					
JCEE	39951	315.34	63.87	134.19	578.17
Percent Male	39951	0.64	0.48	0	1
Age	39951	18.64	0.67	15.67	22.50
Mother's YOS	39925	11.23	3.50	0	18
Father's YOS	39917	10.06	3.54	0	18

TABLE 2—CROSS-VALIDATED RESULTS: FIRST-STAGE AND REDUCED-FORM

A. Probability of National Rank							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(Treat)	0.247*** (0.065)	0.264*** (0.073)	0.279*** (0.076)	0.284*** (0.076)	0.290*** (0.078)	0.287*** (0.078)	0.300*** (0.078)
bwidth	20	40	60	80	100	120	140
order	1	2	3	4	5	6	6
R2	0.60	0.67	0.73	0.76	0.78	0.79	0.79
N	7689	15904	24040	31156	35963	37907	38931
B. Peer Quality @ College							
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
1(Treat)	1.049* (0.497)	2.067*** (0.597)	1.299* (0.636)	2.266*** (0.693)	1.590* (0.690)	1.979*** (0.712)	1.526* (0.708)
bwidth	20	40	60	80	100	120	140
order	1	3	3	5	5	6	6
R2	0.73	0.77	0.81	0.84	0.86	0.87	0.87
N	7689	15904	24040	31156	35963	37907	38931
C. Log Tuition							
	(15)	(16)	(17)	(18)	(19)	(20)	(21)
1(Treat)	-0.165*** (0.047)	-0.177*** (0.051)	-0.183*** (0.055)	-0.187*** (0.054)	-0.190*** (0.056)	-0.188*** (0.055)	-0.196*** (0.055)
bwidth	20	40	60	80	100	120	140
order	1	2	3	4	5	6	6
R2	0.59	0.67	0.74	0.77	0.79	0.80	0.80
N	7689	15904	24040	31156	35963	37907	38931
D. Log Wage							
	(22)	(23)	(24)	(25)	(26)	(27)	(28)
1(Treat)	0.068* (0.031)	0.074* (0.035)	0.076* (0.038)	0.093* (0.043)	0.100* (0.047)	0.108* (0.051)	0.085+ (0.051)
bwidth	20	40	60	80	100	120	140
order	1	2	3	4	5	6	6
R2	0.23	0.22	0.18	0.18	0.18	0.18	0.18
N	1838	4105	6513	8817	10186	10699	10929

Notes: The above regression were run on a model including a treatment indicator for being above or below the admission cutoff, and linear or polynomial function of the running variable, covariates, and cutoff fixed effects. Error term was clustered at the department level. Panel A represents prestige (or strength of first-stage) results for the probability of being admitted into a national college. Panel B is for peer-quality results. Panel C is for reduced tuition. Panel D is the reduced-form benefit in the form of first-year log wages. The results are cross-validated on various bandwidths ("bwidth") and linear/polynomial fits ("order"). Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, *** $p < 0.01$

TABLE 3—CROSS-VALIDATED WAGE RESULTS: FIXED EFFECTS

A. Department Fixed Effects							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(Treat)	0.083* (0.041)	0.101* (0.041)	0.104* (0.044)	0.118* (0.046)	0.113* (0.049)	0.122* (0.052)	0.095+ (0.050)
bwidth	20	40	60	80	100	120	140
order	1	2	3	4	5	6	6
R2	0.34	0.30	0.27	0.26	0.26	0.25	0.25
N	1838	4105	6513	8817	10186	10699	10929
B. College Major Fixed Effects							
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
1(Treat)	0.068* (0.031)	0.074* (0.035)	0.076* (0.038)	0.093* (0.043)	0.100* (0.047)	0.108* (0.051)	0.085+ (0.048)
bwidth	20	40	60	80	100	120	140
order	1	2	3	4	5	6	6
R2	0.23	0.22	0.18	0.18	0.18	0.18	0.18
N	1838	4105	6513	8817	10186	10699	10929
C. Random-Slopes Model for College Majors							
	(15)	(16)	(17)	(18)	(19)	(20)	(21)
1(Treat)	0.088*** (0.031)	0.084* (0.048)	0.090* (0.035)	0.105*** (0.040)	0.105* (0.044)	0.111* (0.048)	0.079+ (0.045)
bwidth	20	40	60	80	100	120	140
order	1	2	3	4	5	6	6
R2	0.27	0.24	0.20	0.20	0.20	0.19	0.19
N	1838	4105	6513	8817	10186	10699	10929

Notes: The tabular results above present the results from regressions of each outcome variable (first-stage and reduced-form) on a model including an indicator for being above or below the admission cutoff, and linear or polynomial function of the running variable, covariates, and fixed effects. The results are cross-validated on various bandwidths ("bwidth") and linear/polynomial fits ("order"). Panel A uses department-level fixed effects, Panel B uses college-major-level fixed effects. Panel C is a random-slopes model, ie, in this model the fixed effects are a college major effects as well as an interaction between each college major and the running variable. Standard errors in parentheses. + p<0.10, * p<0.05, *** p<0.01

TABLE 4—CROSS-VALIDATED RESULTS: RANDOM-SLOPES SOCIAL VS SCIENCE

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	A. Peer Quality @ College											
1(Treat)	0.251 (1.008)	1.936* (0.748)	0.129 (1.014)	1.783* (0.754)	-0.169 (0.999)	1.784* (0.745)	-0.071 (0.977)	1.781* (0.747)	0.047 (0.979)	1.947* (0.768)	0.034 (0.952)	1.962* (0.775)
sample	soc	soc	soc	soc	soc	soc	soc	soc	soc	soc	soc	soc
bwidth	100	100	110	110	120	120	130	130	140	140	150	150
R ²	0.89	0.89	0.89	0.90	0.89	0.90	0.89	0.90	0.89	0.90	0.88	0.90
N	15777	20186	16203	20950	16334	21573	16365	22067	16390	22541	16501	22844
	B. Log Wages											
	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)
1(Treat)	0.080 (0.058)	0.148* (0.072)	0.076 (0.056)	0.118+ (0.068)	0.067 (0.054)	0.102 (0.065)	0.043 (0.053)	0.116+ (0.062)	0.058 (0.051)	0.112+ (0.060)	0.064 (0.048)	0.097+ (0.058)
sample	soc	soc	soc	soc	soc	soc	soc	soc	soc	soc	soc	soc
bwidth	100	100	110	110	120	120	130	130	140	140	150	150
R ²	0.16	0.24	0.16	0.23	0.16	0.23	0.16	0.23	0.16	0.23	0.16	0.22
N	6801	3385	6995	3532	7054	3645	7068	3752	7077	3852	7135	3928

Notes. The tabular results above present the results from regressions of each outcome variable (first-stage and reduced-form) on a random-slopes model (as in Panel C of Table 3) including an indicator for being above or below the admission cutoff, and linear or polynomial function of the running variable, covariates, and fixed effects. The odd-numbered columns are for sub-samples including observations only for social-science college majors in the main sample, that is, business and law majors. The even-numbered columns are for sub-samples including observations only for sciences college majors in the main sample, for example, math, engineer, physical & agricultural sciences, etc. The row labeled ‘sample’ distinguishes between social- and science-majors. The results are cross-validated on various bandwidths (‘bwidth’) and using a 5th order polynomial. Panel A provides first-stage results for peer quality, and Panel B provides reduced-form results for log wages. Standard errors in parentheses. + p<0.10, * p<0.05, *** p<0.01

TABLE 5—CROSS-VALIDATED RESULTS: RANDOM-SLOPES: FEMALE VS. MALE

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
A. Peer Quality @ College												
1(Treat)	1.653* (0.807)	1.499* (0.611)	1.538+ (0.827)	1.302* (0.617)	1.178 (0.810)	1.297* (0.611)	1.815* (0.860)	1.499* (0.636)	1.425+ (0.850)	1.345* (0.629)	1.516+ (0.833)	1.331* (0.625)
sample	female	male	female	male	female	male	female	male	female	male	female	male
bwidth	100	100	110	110	120	120	130	130	140	140	150	150
R ²	0.88	0.89	0.88	0.90	0.88	0.90	0.88	0.90	0.88	0.90	0.88	0.90
N	13171	22792	13564	23589	13772	24135	13909	24523	14037	24894	14205	25140
B. Log Wages												
1(Treat)	0.102* (0.047)	0.092 (0.100)	0.097* (0.045)	0.048 (0.098)	0.075+ (0.044)	0.037 (0.095)	0.106* (0.047)	0.070 (0.103)	0.095* (0.046)	0.049 (0.101)	0.097* (0.045)	0.060 (0.098)
sample	female	male	female	male	female	male	female	male	female	male	female	male
bwidth	100	100	110	110	120	120	130	130	140	140	150	150
R ²	0.10	0.05	0.11	0.05	0.11	0.05	0.11	0.05	0.11	0.05	0.11	0.05
N	7313	2873	7544	2983	7643	3056	7708	3112	7760	3169	7853	3210

Notes. The tabular results above present the results from regressions of each outcome variable (first-stage and reduced-form) on a random-slopes model (as in Panel C of Table 3) including an indicator for being above or below the admission cutoff, and linear or polynomial function of the running variable, covariates, and fixed effects. The odd-numbered columns are for results including only females, while even-numbered columns are for males—as list in the ‘sample’ row. The results are cross-validated on various bandwidths (‘bwidth’) and using a 5th-order polynomial until 120 points either side and then 6th-order afterwards. Panel A provides first-stage results for peer quality, and Panel B provides reduced-form results for log wages. Standard errors in parentheses. + p<0.10, * p<0.05, *** p<0.01

TABLE 6—CROSS-VALIDATED RESULTS: RANDOM-SLOPES: TRACK AND GENDER INTERACTED

	A. Peer Quality						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(Treat)	1.015* (0.441)	1.424*** (0.497)	1.950*** (0.552)	1.174* (0.555)	0.084+ (0.049)	0.100* (0.043)	0.105*** (0.040)
bwidth	20	30	40	50	60	70	80
order	1	2	3	3	4	4	4
R2	0.81	0.81	0.82	0.85	0.86	0.86	0.87
N	7689	11620	15904	20181	24040	27765	31156
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
1(Treat)	0.096* (0.038)	1.629*** (0.623)	1.427* (0.634)	1.276* (0.627)	1.646* (0.656)	1.414* (0.650)	1.438* (0.642)
bwidth	90	100	110	120	130	140	150
order	4	5	5	5	6	6	6
R2	0.88	0.89	0.89	0.89	0.89	0.89	0.89
N	34171	35963	37153	37907	38432	38931	39345
	B. Log Wages						
	(15)	(16)	(17)	(18)	(19)	(20)	(21)
1(Treat)	0.088*** (0.031)	0.089* (0.040)	0.082+ (0.048)	0.070+ (0.041)	0.084+ (0.049)	0.100* (0.043)	0.105*** (0.040)
bwidth	20	30	40	50	60	70	80
order	1	2	3	3	4	4	4
R2	0.27	0.24	0.24	0.21	0.20	0.20	0.20
N	1838	2917	4105	5307	6513	7722	8817
	(22)	(23)	(24)	(25)	(26)	(27)	(28)
1(Treat)	0.096* (0.038)	0.106* (0.044)	0.090* (0.042)	0.074+ (0.041)	0.093* (0.046)	0.079+ (0.045)	0.084+ (0.043)
bwidth	90	100	110	120	130	140	150
order	4	5	5	5	6	6	6
R2	0.20	0.20	0.19	0.19	0.19	0.19	0.19
N	9656	10186	10527	10699	10820	10929	11063

Notes: The tabular results above present the results from regressions of each outcome variable (first-stage and reduced-form) on a random-slopes model (as in Panel C of Table 3) including an indicator for being above or below the admission cutoff, and linear or polynomial function of the running variable, covariates, and fixed effects. The results are cross-validated on various bandwidths ("bwidth") and orders of polynomial ("order"). Panel A provides first-stage results for peer quality, and Panel B provides reduced-form results for log wages. Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, *** $p < 0.01$

For Online Publication

Early Admissions

Early admissions was introduced in 1994 and has steadily expanded in the shares of applicants participating and applicants accepted. In 2000, 57,820 high school seniors applied to college through early admissions (47 percent of all college applicants); of which just over 7039 or 13.5 percent were admitted. The set up of the early admissions process works as follows: starting in early February (before regular admissions), any high school senior may sit for a set of 5 subject-specific tests, referred to as the General Standardized Achievement Test (here after “the February exams”).¹ Once students receive their scores for the February exams, they make their single choice of where to apply. Unlike regular admissions, early admissions allows students only *one* chance to apply *directly* to *one* specific department. Early admissions decisions are left primarily to the department’s admissions committee, who are closely supervised by the MOE.

After the February exams, applications are reviewed by departments, and short-listed applicants are interviewed by the department’s admissions committee. Offers are made to the applicants; there are always more applicants that a department would like to admit than there are slots available during this round. At this point, students must choose to accept or reject the admissions offer. In order to participate in the regular admissions, students must not be holding an early admissions seat. All unfilled seats must be in play during the regular admissions process in order for the matching mechanism to be efficient. Those who compete in early admissions but fail must then compete in regular admissions in order to be admitted at all.

How Colleges Rank Students

¹The February exams differ from the JCEE in two ways: first, all 5 subject-tests are required for the February exams; second, the February exams test general knowledge of the type learned in high school, unlike the JCEE which tests for track-specific abilities needed in college.

Let T_{id} represent the score of student i being considered by the matching algorithm for admissions to department d . T_{id} does not necessarily equal the simple summation of raw scores on all the subject tests that a student may have taken on the JCEE. Instead, it is the combined realization of choices made by the students, the MOE, and the departments. More precisely, T_{id} is a weighted exam score; it provides the priority ordering to the matching algorithm and corresponds to the running variable in a RDD.

The JCEE is a set J of 9 subject-specific tests written and administered by the MOE, one subject test t^j for each subject $j \in J$. Of the 9 subject tests only 5 to 6 are required by the MOE for admission. (Panel A of Figure 2 provides the density plot of these scores, which approximate a normal distribution.) The MOE determines which subject tests are required for admission to any given department, according to the educational track the department falls into. Departments have no control over whether a subject j is required or not. Thus, the dummy ρ_d^j in the formula above indicates (1=yes/0=no) whether or not a specific j is required for that specific department d , which offers a specific college major m according to the MOE's set of college majors M .

Even though departments have no influence over required subject tests, the MOE does allow departments two ways to indirectly exercise their priorities for student quality: choosing a set of weights w_d^j and a set of subject-specific raw-score cutoffs κ_d^j to be applied to subject-specific test scores t_i^j from the JCEE. First, the MOE requires departments to choose weights w_d^j that increase the value of that required subject relative to other ones, allowing the department to emphasize that specific content knowledge over other content when summing them together. The menu of w consists of 1, 1.25, 1.5, 1.75, or 2. The default and most common weight is $w=1$, which leaves the raw score for that required subject score unchanged.²

²One could replace the ρ above with a $w=0$. But since departments have influence over which weights (w) to use but do not have influence over which subject tests are required (ρ), we choose to keep these features distinct in the model.

Second, the MOE requires departments to choose from a menu of subject-specific raw-score cutoffs κ_d^j to use on subject-specific raw scores t_i^j . Raw-score cutoffs let departments truncate the distribution of viable candidates at some point in the distribution of t_i^j for each required subject test. Departments are not allowed to set where the cutoffs occur. The same four κ^j are available for any of the required subjects j . The default and most common raw-score cutoff is $\kappa^j=0$, which leaves the distribution of applicants unchanged. Specifically, the four κ^j are:

- A: only consider students with t_i^j above the average of the upper 50th percentiles for subject j ;
- B: only consider students with t_i^j above the 50th percentile for subject j ;
- C: only consider students with t_i^j above the average of the lower 50th percentiles for subject j ; or
- D: no raw score cutoff, which is analogous to $\kappa^j=0$.

Any student with a raw score t_i^j below the raw-score cutoff κ_d^j are automatically disqualified. In the data replication process we follow to generate the running variable around each national-college cutoff, if the condition $t_i^j \geq \kappa_d^j$ is not met, we approximate the algorithm's disqualification by giving i a missing value $\emptyset$ for T_{id} for that department.

Based on the above description, the formula for calculating T_{id} for each academic department is:

$$T_{id} = \begin{cases} \sum_{i=1}^N \sum_{d=1}^m \sum_{j=1}^J t_i^j \rho_d^j w_d^j & \text{if } t_i^j \geq \kappa_d^j \forall j \\ \emptyset & \text{otherwise.} \end{cases}$$

Notice that the formula above replicates counterfactuals only within college majors; for example, a student who is admitted into a linguistics department would have priority rankings replicated for her at other linguistics departments, but not a history department. Since we estimate the returns to attending a

national college versus a private college within the same major, we replicate only one counterfactual per college major. It is the counterfactual associated with the cutoff for admissions into the academic department in the lowest ranked national college; that is, the cutoff is the lowest test score of any student admitted into a national college for that college major. Thus, the running variable for that college major is normalized around the weighted test scores for that department's cutoff, generated by their weights and raw-score cutoffs. Even though students could apply to more than one college major, we limit the counterfactuals we inspect in this way so that we can reasonably compare the monetary outcomes for those with the same college major.^{3,4}

To reconstruct T_{id} as it was generated by the algorithm, we use the MOE's records of department priorities for 2000, which detail how each department chooses to use weights and raw-score cutoffs. An example of a hypothetical matching process is provided below.

Use of weighted and unweighted scores in analysis

In constructing the running variable for the RD, we use the weighted scores T_{id} because the algorithm uses the weighted score to sort students into priority ordering for seats, and thus the use of the weighted scores faithfully reconstructs the cutscore and the relative distance to the cutscore that each student would have experienced while the algorithm was allocating seats. However, the weights themselves have no substantive or economic meaning outside the allocation process, because it was the raw scores that students were given on the JCEE. Indeed, the weights are specific to the departments, and so it would be difficult to make comparisons of and interpret the difference in weighted scores across departments. Since subject-specific test scores t_i are the JCEE's measure of student perfor-

³Unlike ?, who replicate the counterfactuals for each student around multiple discontinuities, this analysis considers only one of many possible counterfactuals for each applicant; therefore, the data is never duplicated.

⁴In theory, we could replicate the counterfactuals for each student at every department within the specific college major where she was actually enrolled because we have the data to do this; but this would provide a different focus to the paper.

mance, we use their unweighted total sum T_i^u (corresponding to the individual's actual score) to construct the average peer quality $\overline{A_i}$ for each academic department.

Matching example

To illustrate how required subject tests (ρ^j), weights (w^j), and raw-score cutoffs (κ^j) work in defining the RD's running variable, take the hypothetical example with four students i_1 , i_2 , i_3 , and i_4 . Each is competing for the last seat in the same department. In this hypothetical example, the department's priorities for student quality lead it to value $t_i^{chinese}$ 1.5 times that of t_i^{math} and to want only students with a $t_i^{chinese}$ above $\kappa_d^{chinese(A)}$, that is, above the average of the top 50th percentiles in j =chinese.⁵ Let us assume for the purposes of this hypothetical example that $\kappa_d^{chinese(A)}$ =54 and that these two subjects are the only ones required for the matching to take place.

Looking at the matrix below, we can see that i_4 is automatically disqualified because he did not take the required math subject test. Next, we see that i_3 dominates in both raw and weighted total scores. However, because i_3 's Chinese score is below the threshold $\kappa^{chinese(A)}$ for consideration, i_3 is also not admitted to d .

	$t_i^{chinese}$	t_i^{math}	Raw Total	Weighted Total	Enrolled?
i_1	64	66	131	162	yes
i_2	54	80	134	161	no
i_3	52	100	152	$178 \implies \emptyset$	no
i_4	88	—	88	$\emptyset$	no

Both i_1 and i_2 have $t_i^{chinese}$ above the raw-score threshold; and i_2 has the advantage in total raw score. However, because the department's priority weights Chinese by 1.5, i_1 narrowly beats i_2 for the seat. Once again, this is because i_1 's

⁵This means that $w^{chinese}$ =1.5, w^{math} =1, $\kappa^{chinese}$ =A (or 54), and κ^{math} =D (or zero).

total weighted score is higher, not because i_1 's $t_1^{chinese}$ is highest. Had i_2 been able to score 1.5 more points either by increasing her $t_2^{chinese}$ by 1 point or t_2^{math} by 2, she would have bested i_1 for the seat.

Labor Data

Despite our preference for a complete match, possessing one-third of the complete universe of college graduates from the 2000 cohort is still a considerable advantage over most analyses that rely on census or survey data. These studies rely on selected subsamples of the whole population; thus, if we were to focus in those data solely on the graduates from the 2000 cohort, the number of observations that would be available are considerably fewer. Therefore, we take the 33 percent match rate and the evidence for no selection in or out of the Labor data as positive contributions. We do run a separate regression for women only to test for sensitivity. This substantially reduces the sample size in these regressions, so the results will be noisier.

The Labor data we have access to is taken from the employee earnings records for the month of December for each year. There are advantages to this feature of the data. The first advantage of having monthly earnings from December is that we can capture earnings soon after graduates complete their education in June. A second advantage is that the data is not self-reported, but rather is supplied by the employers to the central government, as part of uniform and routine administrative processes. Self-reported earnings, which are common in survey data, have been shown to be less precise measures of actual earnings. A third advantage over monthly earning records in survey data is that we record all observations on the same month of the year, which protects against the noise induced by business-cycle seasonality in monthly earnings. A possible disadvantage, however, is that if a college graduate were between jobs in either December 2004 or December 2005, the data would not capture his/her employment. The only way to remedy this would be to have the monthly earnings for every month. Unfortunately, it is

politically impossible to obtain this extent of data.

First-year earnings, as opposed to earnings a few years after graduation, is a theoretically appealing measure for this study because it would reflect an employer's reliance on college quality as a signal of students' potential productivity. If employers use college quality as a signal to predict an applicant's expected productivity, then first-year earnings should follow a similar pattern as the variance in the signals of college quality the employers observe(?). Another way first-year earnings could be useful in this study is as a proxy for longer-term earnings. ? show that a college graduate's starting salary is predictive of long-term earnings growth; thus, on average, starting off on a lower (higher) rung of the pay scale means that longer term earnings will also be lower (higher).